Identity 3.13: $\cos 3A = 4\cos^3 A - 3\cos A$

Proof.

$$\begin{aligned}\text{We have, } \cos 3A &= \cos(2A + A) = \cos 2A\cos A - \sin 2A\sin A \\ &= \left(2\cos^2 A - 1\right)\cos A - 2\sin A\cos A\sin A \\ &= \left(2\cos^2 A - 1\right)\cos A - 2\cos A\left(1 - \cos^2 A\right) \\ &= 4\cos^3 A - 3\cos A.\end{aligned}$$

Identity 3.14: $\tan 3A = \dfrac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$

Proof.

$$\begin{aligned}\text{We have, } \tan 3A &= \tan(2A + A) \\ &= \frac{\tan 2A + \tan A}{1 - \tan 2A\tan A} \\ &= \frac{\dfrac{2\tan A}{1 - \tan^2 A} + \tan A}{1 - \dfrac{2\tan A}{1 - \tan^2 A}\tan A} \\ &= \frac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}.\end{aligned}$$

Double and Triple angle identities are given below:

sine	cosine	Tangent
$\sin 2A = 2\sin A\cos A$	$\cos 2A = \cos^2 A - \sin^2 A$	$\tan 2A = \dfrac{2\tan A}{1 - \tan^2 A}$
$\sin 2A = \dfrac{2\tan A}{1 + \tan^2 A}$	$\cos 2A = 2\cos^2 A - 1$	$\tan 3A = \dfrac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$
$\sin 3A = 3\sin A - 4\sin^3 A$	$\cos 2A = 1 - 2\sin^2 A$	
	$\cos 2A = \dfrac{1 - \tan^2 A}{1 + \tan^2 A}$	
	$\cos 3A = 4\cos^3 A - 3\cos A$	

Half-Angle Identities

Half angle identities are closely related to the double angle identities. We can use half angle identities when we have an angle that is half the size of a special angle. For example, $\sin 15^\circ$ can be computed by writing $\sin 15^\circ = \sin\dfrac{30^\circ}{2}$. Also one can find exact values for some angles using half-angle identities. If we put $2A = \theta$ or $A = \dfrac{\theta}{2}$ in the double angle identities, we get new identities in terms of angle $\dfrac{\theta}{2}$.